SENG 310 ASSIGNMENT 4

MEDIUM-FIDELITY PROTOTYPING AND EVALUATION PLAN

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SENG 310 – Human Computer Interaction

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1. DESCRIPTION OF THE PROTOTYPE

1.1 Prototype Overview

The Palendar medium-fidelity prototype is an interactive web-based application designed to address the challenge of coordinating social activities across multiple friend groups with conflicting schedules. The prototype demonstrates the complete user journey from authentication through event creation, implementing the core functionality required for users to identify common free times and schedule group hangouts efficiently.

The prototype encompasses six primary interface sections: (1) Login/Authentication, (2) Home Dashboard, (3) Personal Calendar, (4) User Profile, (5) Groups Management, and (6) Friends Chat. Each section has been designed with deliberate attention to user flow, visual hierarchy, and task completion efficiency based on insights gathered from our PA2 user research with eight university students.

1.2 Meeting Design Requirements from PA2

Our prototype successfully addresses all five "must be met" design requirements identified in PA2:

Requirement 1: Ability to create and view personal calendar
The prototype implements a comprehensive personal calendar interface accessible from the home dashboard. Users can view their schedules with clear visual differentiation between personal events and group events. Days with universal availability across all groups are highlighted with green titles, directly addressing Vidit's need for "viewing all components separately in schedules" (PA2). The calendar provides both weekly and event-based views, allowing users to manage time commitments across academic, work, and social domains with color-coded categories as suggested by Abheenaya during interviews.

Requirement 2: Ability to join multiple groups
The Groups section enables users to participate in unlimited friend groups simultaneously. The interface displays all groups in a scrollable list format with visual indicators for each group's status. Users can join groups by entering a group code through the "Join Group" function, directly addressing Abheenaya's challenge of "managing time across multiple groups" while preventing the double-booking issues she experienced with existing tools.

Requirement 3: Ability to view a group's calendar
Clicking on any group navigates users to a dedicated group calendar view that displays the aggregated schedules of all group members. This feature fulfills Harshad's request for "a clear way to see when friends are busy or free, all in one shared calendar" to prevent the last-minute conflicts that frequently caused his restaurant outings to be canceled (PA2).

Requirement 4: Group informs members about common free times
The prototype implements a visual indicator system using green icons to denote time slots

when all group members are available. When users click on these indicators, the interface displays explicit information about availability: "times when everyone is free, times when some are free, or most are free." This automated identification addresses Apoorva's pain point that "hangout plans fail very frequently because people have different class times" and eliminates the need to manually compare individual schedules (PA2).

Requirement 5: Ability to create a group
The "Create Group" functionality allows users to establish new groups by specifying a group name, selecting a category type, and providing a description. The system then generates a unique group code that can be shared with friends for joining, implementing the nine-digit code system proposed in PA1.

Should Be Met Requirements Implemented:
The prototype also incorporates one of the "should be met if possible" requirements:

Group chat functionality: The Friends tab provides an integrated chat interface where users can communicate and coordinate events, making Palendar "an end-to-end solution for facilitating planning of events within groups" as specified in our PA2 design requirements. This addresses interview feedback about the fragmentation between scheduling tools and communication platforms.

1.3 Changes Based on PA3 Heuristic Evaluation

Our PA3 heuristic evaluation identified several usability issues that informed critical design improvements in this medium-fidelity prototype:

Addressing Visibility of System Status (Severity 2 issue from PA3):
The heuristic evaluation revealed insufficient feedback when users performed actions. The current prototype addresses this through explicit visual indicators: green icons clearly show group availability status, event creation forms provide immediate confirmation, and navigation elements highlight the current section. The profile page includes a "Recent Activity" section that provides users with system status awareness.

Resolving Color-Coding Confusion:
PA3 identified potential confusion with the color-coding system. The revised prototype implements a more intuitive approach: green universally indicates availability/positive states (free times, available groups), while the calendar uses distinct color categories for personal versus group events. Days with universal group availability are marked with green titles, creating consistent color semantics across the interface.

Reducing Interface Clutter:
The evaluation noted that earlier sketches (particularly the group calendar sketch by Sam) contained "too many buttons and random features that made the UI cluttered and messy" (PA3). Following our team's debate on aesthetics versus usability informed by Tuch et al. (2012) and Hassenzahl (2004), we implemented a minimalist approach. The group interface focuses on essential actions—view calendar, create event, and manage groups—with

secondary functions accessible through clear but unobtrusive buttons.

Improving Recognition Over Recall:
The prototype provides persistent navigation elements and clear labeling throughout. The home dashboard presents all primary functions (Calendar, Profile, Events) upfront, reducing memory load. Event creation forms use familiar input patterns with explicit labels, and the groups list displays group names and categories prominently to aid recognition.

1.4 Design Rationale and Justification

Task-Centered Design Approach:
Our design decisions directly implement the task descriptions developed in PA2 using the Task-Centered System Design (TCSD) methodology. The primary task flow—login → select group → identify free times → create hangout—aligns precisely with our personas' needs, particularly "Busy University Student Maya" who struggles with scheduling across multiple commitments.

Visual Hierarchy and Information Architecture:
The three-button home dashboard (Calendar, Profile, Events) emerged from our PA3 brainstorming "separation of concerns approach" (Component 2 analysis). This design prevents overwhelming users with information while providing direct access to core functions. Research on visual hierarchy in interface design supports this approach, demonstrating that users navigate more efficiently when primary actions are visually prominent (Wickens et al., 2013).

Color Semantics for Availability:
The use of green to indicate universal availability follows established conventions in scheduling interfaces and traffic light metaphors familiar to users. This design choice emerged from our PA2 interview with Abheenaya, who specifically suggested color-coding schedules, and was refined through our PA3 analysis to ensure accessibility and clarity.

Scrollable Group Lists:
Our decision to implement scrolling rather than pagination for group displays (PA3 Component 3 analysis) accommodates users like Abheenaya who manage multiple friend groups. This "combination of both" approaches provides flexibility while maintaining visual organization.

Integrated Communication:
Including the Friends chat functionality reflects our understanding from PA2 interviews that scheduling and communication are intertwined activities. Harshad mentioned relying on "messy group chats" for coordination—by integrating messaging, Palendar reduces app-switching and maintains context within the scheduling workflow.

Form Design for Event Creation:
The event creation form balances completeness with simplicity, requesting only essential information (name, location, date, attendees). This design emerged from our PA3 discussion

on the Create Group page (Component 6), where we identified that straightforward features benefit from minimal interface elements. The form follows Gestalt principles of grouping related fields and uses familiar input patterns to reduce learning curve.

2. EVALUATION PLAN

a. Goal for the Usability Testing User-Evaluation

The primary goal of this usability evaluation is to assess whether users can successfully identify common free times among group members and create a group hangout using the Palendar interface, while measuring overall usability, efficiency, and user satisfaction with the medium-fidelity prototype.

Specific objectives include:
1. Validating that the prototype meets the "must be met" design requirements from PA2, particularly the group calendar visualization and common free time identification
2. Assessing whether improvements made based on PA3's heuristic evaluation have resolved identified usability issues (visibility of system status, color-coding clarity, interface clutter)
3. Measuring task completion success rates, time-to-completion, and error frequencies for the core user workflow
4. Gathering qualitative feedback on interface clarity, satisfaction, and likelihood of adoption
5. Identifying any remaining usability problems that require resolution before final implementation

Success will be determined by achieving 75% task completion rates, average satisfaction ratings of 4/5 or higher, and users expressing that Palendar improves upon their current scheduling methods.

b. Steps for a Single Evaluation Session

Each 35-40 minute evaluation session follows a structured eight-phase protocol:

Phase 1: Pre-Session Setup (5 minutes)
Before participant arrival, load the prototype at the login screen, prepare screen recording software (OBS Studio), set up audio recording on a smartphone, print consent forms and scenario cards, prepare note-taking materials, test all recording equipment, and silence device notifications. Upon participant arrival, greet them warmly, review and sign the consent form emphasizing voluntary participation and anonymity, and explain that we are testing the interface rather than their abilities.

Phase 2: Pre-Task Interview (5 minutes)
Collect demographic information (year, program, undergraduate/graduate status), current scheduling practices ("How do you plan hangouts with friends? What tools do you use?"), pain points ("What frustrates you most about scheduling group activities?"), and technology comfort level (1-5 scale, current calendar app usage). Document responses for participant characterization and later analysis.

Phase 3: Think-Aloud Training (3 minutes)
Explain the think-aloud protocol: "As you use the prototype, speak your thoughts aloud—what

you're looking for, what you expect, what confuses you." Demonstrate using an unrelated example (weather app), have the participant practice briefly, and emphasize that we cannot provide assistance during tasks, but getting stuck provides valuable feedback.

Phase 4: Task Scenario Introduction (2 minutes)
Present a printed scenario card: "You and your friend group want to plan a movie night this week. Your group 'SENG 310 Squad' has already added schedules to Palendar. Your task: (1) Log in, (2) Navigate to your group, (3) Find when all 4 members are free, (4) Create a 'Movie Night' hangout for Thursday 7-9pm." Verify understanding, then start screen and audio recordings.

Phase 5: Task Execution (15-20 minutes)
Observe participants completing four sub-tasks:

Task 1—Login (1-2 min): Note which login method they choose, any hesitation, completion time. If stuck >1 minute, ask "What are you expecting?" then provide gentle hint if needed.

Task 2—Navigate to Group (1-2 min): Observe if they find Groups section intuitively, click correct group, understand page layout. If stuck >45 seconds, prompt: "Where would you look for friend groups?"

Task 3—Identify Common Free Times (2-4 min): Watch how they interpret the calendar, color-coding, and availability indicators. Ask probing questions: "What do these colors mean?" "How can you tell when everyone is free?" If stuck >90 seconds, hint toward the visual indicator system.

Task 4—Create Hangout (2-4 min): Note if they find the create button, understand form fields, successfully input details. If stuck >1 minute, ask: "What action would schedule this time?" Document completion status, time, and errors for each sub-task.

Phase 6: Post-Task Interview (8-10 minutes)
Gather immediate reactions ("What are your overall thoughts?"), specific feature feedback (calendar clarity, color-coding effectiveness, common free times usefulness), task-specific reflections (ease of login, navigation, calendar interpretation, event creation), and comparative analysis ("How does this compare to your current methods? Would you use this app?"). Administer quantitative ratings (5-point Likert scales): task ease, interface satisfaction, likelihood of use, confidence in independent use, calendar visualization helpfulness. Conclude with improvement suggestions: "If you could change one thing, what would it be? Any missing features?"

Phase 7: Session Wrap-Up (2 minutes)
Thank participants for valuable feedback, answer any questions about the study or app, remind about data confidentiality and anonymous reporting, provide team contact information, and stop all recordings.

Phase 8: Post-Session Documentation (5-10 minutes)
Immediately type organized notes while session is fresh, rename recording files with participant ID and date (P03_Nov07_screen.mp4), complete session summary template (participant demographics, task completion status, notable quotes, critical incidents, severity ratings), and input quantitative data into master spreadsheet.

c. Anticipated Problems and Challenges

Prototype Limitations: The medium-fidelity prototype may have limited interactivity—not all buttons respond, unexpected user paths may not be supported, and transitions may feel unnatural. Users with color vision deficiencies may struggle with the color-coding system despite improvements from PA3.

Participant-Related Issues: Think-aloud verbalization feels unnatural for many users, especially when concentrating on complex tasks. Technology experience varies significantly—PA2's Shreya represented users uncomfortable with technology, while others may be highly technical. Social desirability bias may cause participants (especially friends/acquaintances) to soften criticism or overstate willingness to use the app.

Methodological Challenges: Time constraints may cause rushed sessions or incomplete post-task interviews if participants take longer than expected. Different team members facilitating sessions may provide inconsistent hints or vary question phrasing. Recording equipment could fail (battery death, storage full, software crash). Calendar visualization complexity may overwhelm some users, requiring significant time to parse information.

Logistical Issues: Recruiting 4-8 participants from our network with appropriate availability is challenging. Last-minute cancellations could disrupt our testing timeline. Maintaining participant anonymity is difficult if they know each other. Analyzing large amounts of qualitative data (recordings, notes, open-ended responses) and determining which issues represent genuine usability problems versus individual quirks requires careful judgment.

d. Remedies for Those Problems

For Prototype Limitations: Test all interactive elements thoroughly before sessions and create a "contingency script" for non-functional areas ("In the full app, that would [describe action]"). Add a clear legend explaining color meanings and use color + pattern combinations (stripes, dots) to support color-blind users, following WCAG accessibility guidelines. Design a clear "happy path" through the prototype that minimizes branching.

For Participant Issues: Provide thorough think-aloud training with practice examples and use gentle prompts during silence: "What are you thinking right now?" Screen participants during recruitment to target moderate tech comfort (3-4/5). For low-tech users, note their experience level and potentially weight feedback differently. Emphasize repeatedly that negative feedback is valuable and frame questions neutrally: "What worked and what didn't?" rather than "Did you like it?" Have team members unknown to participants serve as facilitators when possible to reduce social pressure.

For Methodological Challenges: Build 10-minute buffers between sessions for overruns. If participants get stuck >3 minutes, provide strong hints or demonstrate to keep progressing. Use two recording devices (computer + phone backup). Create a detailed facilitation script that all team members follow verbatim. Conduct a pilot session together where all members observe to calibrate approach and hint-giving. Test equipment before each session (battery levels, storage space, microphone functionality).

For Calendar Complexity: Implement a prominent "Common Free Times" summary box listing available slots in text format. During testing, specifically probe understanding: "Can you explain what you're seeing?" If >2 minutes are consistently spent parsing the calendar, note it as a critical redesign priority.

For Logistical Issues: Over-recruit 10-12 potential participants to account for drop-outs. Provide flexible scheduling (morning/afternoon/evening slots). Send 24-hour reminder messages. Create a standardized data template and input data immediately after each session. Use affinity diagramming to organize qualitative findings thematically. Focus on patterns affecting 3+ participants rather than individual outliers. Assign participant ID numbers (P01, P02) and use only those in documentation. Review recordings before sharing to redact any names mentioned.

e. Data Collection

Quantitative Metrics:
Task performance data includes completion rates for each sub-task (login, navigate to group, identify free times, create hangout) recorded as success/failure/completed-with-hints, task completion times measured in minutes:seconds, number and types of errors (navigation, input, conceptual), and number of facilitator hints required. Post-task Likert scale ratings (1-5) assess task ease, interface satisfaction, likelihood of regular use, confidence in independent use, and calendar visualization helpfulness. Demographic data captures year, program, technology comfort level, current calendar app usage, and number of friend groups.

Qualitative Data:
Think-aloud protocol captures real-time verbalizations, spontaneous comments, tone/emotion, and problem-solving thought processes via audio recording. Facilitator observation notes document interaction patterns (where users click first, exploration vs. direct navigation), hesitations and pauses indicating confusion, facial expressions and body language, navigation paths and backtracking, error recovery strategies, and unexpected behaviors. Semi-structured interviews gather information on current scheduling methods, specific pain points, feature-specific feedback (calendar clarity, color-coding, common free times display), comparative analysis with existing tools, improvement suggestions, and feature expectations.

Recording and Documentation:
Screen recordings capture complete interaction sequences, mouse movements, click patterns, exact navigation paths, and visual evidence of errors and successes using OBS Studio at 1080p, 30fps. Audio recordings preserve think-aloud narration and interview responses. After each session, facilitators complete standardized summary forms

documenting participant ID/demographics, task completion status, time metrics, 3-5 notable quotes, critical incidents, severity ratings for issues, and overall performance assessment.

Data Organization and Analysis:
A master spreadsheet consolidates quantitative data with rows for participants and columns for all metrics, enabling calculation of averages, success percentages, and standard deviations. Affinity diagramming organizes qualitative observations into themes (Calendar Visualization, Color Coding, Navigation Flow). A usability issue log systematically tracks problem descriptions, number of participants affected, location of occurrence, severity ratings (0-4 scale from Nielsen), and recommended fixes. All data is stored in password-protected folders with participant P-numbers only, following ethics approval requirements.

Analysis involves calculating means and standard deviations for rating scales, determining task completion percentages, comparing times across tasks to identify bottlenecks, transcribing key audio portions for critical quotes, coding data thematically, and triangulating quantitative ratings with qualitative feedback to identify priority issues. Findings are organized by task and design requirement, with issues prioritized by severity and frequency for PA5 reporting.

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